Cavernous Sinus Thrombosis

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KEYWORDS

• Cavernous sinus • Thrombosis • Venous sinus thrombosis • Cavernous sinus syndrome • Danger triangle

KEY POINTS

- Cavernous sinus thrombosis is a rare but potentially lethal disease.
- Patients with this disease can present with meningitic symptoms, proptosis, chemosis, pyrexia, cranial nerve deficits, papilledema, vision loss, severe lethargy, and/or coma.
- Cavernous sinus thrombosis may occur secondary to septic and aseptic etiologies.
- Risk factors include infection of the "danger triangle," neoplasia, hormone-replacement therapy, pregnancy, thrombophilic disorders, and trauma.
- Treatment depends on the etiology and symptomatology and may involve antibiotics, anticoagulation, corticosteroids, and surgery.

INTRODUCTION

Cavernous sinus thrombosis (CST), a subset of cerebral sinus thrombosis, is a rare, but potentially life-threatening, condition that involves the formation of a thrombus within the cavernous sinus (CS). The annual incidence of CST is estimated to be 0.2 to 1.6 per 100,000 persons.[1] Although cerebral sinus thrombosis generally exhibits a female predominance, with up to 75% of those affected being women in the adult population and a near-equal gender distribution in pediatric and geriatric populations, CST appears to have an overall slight male predominance.[2,3] Because of its infrequent presentation, prognostication can be difficult; however, mortality rates have steadily declined with time. Recent estimates of morbidity and mortality are approximately 15% and 11%, respectively.[3]

The clinical features of CST are related to the unique anatomy and physiology of the CS, which contains portions of the internal carotid artery, oculomotor nerve (cranial nerve [CN] III), trochlear nerve (CN IV), trigeminal nerve (CN V1 and V2), and the abducens nerve (CN VI).[4,5] Additionally, the CS collects venous blood from the superior and inferior ophthalmic veins, sphenoid sinus, and the superficial cortical vein before draining via the superior and inferior petrosal sinuses and the pterygoid venous plexus (**Fig. 1**).[4] Communication between the bilateral CSs occurs between the anterior, inferior, and posterior interCSs and the basilar sinus.[5] Because of these extensive interconnections, symptoms associated with CST vary and include headache, pyrexia, ocular pain, proptosis, chemosis, periorbital edema, and CN palsies secondary to direct compression or irritation within the CS.[6] Clinicians must remain vigilant in diagnosing CST because it has the potential to rapidly become lethal.

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Neurosurg Clin N Am 35 (2024) 305–310
https://doi.org/10.1016/j.nec.2024.02.002
1042-3680/24/

neurosurgery.theclinics.com

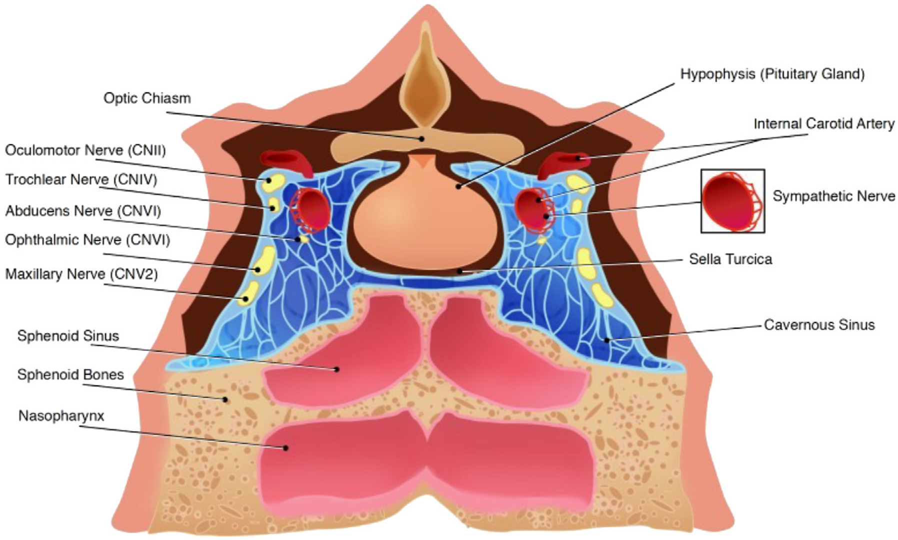

Fig. 1. Anatomy of the cavernous sinus. CN, cranial nerve. ("File:Anatomy of the cavernous sinus.jpg" by Okkes Kuybu, MD and Diana is licensed under CC BY 4.0. To view a copy of this license, visit https://creativecommons.org/licenses/by/4.0/?ref=openverse.)

DISCUSSION
Etiology

CST most commonly results from septic etiologies. These include sinusitis, otitis, erysipelas, mastoiditis, and odontogenic and facial infections.[7] In fact, most cases (60%–80%) arise from infection of the middle third of the face in what is known as the "danger triangle"[5,8] (**Fig. 2**). Ethmoid and sphenoid sinusitis represent more than 50% of the sources, middle ear infections are estimated to cause 10%, and both oral and dental infections have been estimated to represent less than 10% of cases.[5] *Staphylococcus aureus* is the most common microorganism, accounting for approximately 66% to 70% of cases.[5,7] *Streptococcus* species (20%), oral anaerobic flora (10%), and gram-negative bacteria (5%) may also be involved.[5] Less commonly, fungal etiologies, such as aspergillosis, mucormycosis, and actinomycosis, and/or viral etiologies have been reported.[5,9]

Despite the preponderance of septic causes, aseptic cases do occur. For instance, trauma and surgery to the head and neck region may lead to CST. Thrombophilic disorders, such as factor V Leiden mutation, myeloproliferative disorders, antiphospholipid syndrome, prothrombin G20210 A mutation, hyperhomocysteinemia, protein C and/or S resistance or deficiencies, and antithrombin deficiencies, have also been associated with CST.[2] Neoplasia, both local and metastatic, has been reported. Tumors in the CS region are most commonly cancers of the head and neck as well as pituitary adenomas, cavernous meningiomas, cavernous hemangiomas, and neurogenic tumors of the cranial CNs within the CS. More uncommonly, metastases from the breast and lung and prostate cancer have been reported.[9,10] Lymphomas have also

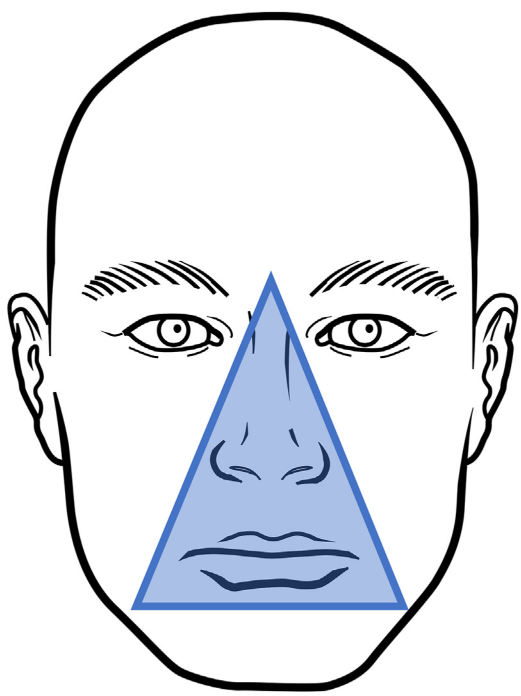

Fig. 2. Diagram illustrating the "danger triangle" of the face. (*Courtesy* of Paul H. Dressel BFA) (no permission required)

been reported to be involved.[11–14] Finally, hormone-based oral contraceptives, hormone-replacement therapies, and pregnancy have also been associated with CST.[2,5,7,15,16]

Differential Diagnosis

Early in the course of CST, nonspecific symptoms, such as pyrexia, headache, emesis, and signs of meningeal irritation, may lead to a mistaken diagnosis of meningitis. Because CST and meningitis often coexist, appropriate evaluation and potential treatment of meningitis should be performed.[5,17] In septic CST, lumbar puncture will often demonstrate an inflammatory profile, yet cerebrospinal fluid cultures are negative approximately 20% of the time.[5]

Orbital cellulitis and superior orbital fissure syndrome may both present in a very similar manner to CST. Patients who have orbital cellulitis may experience periorbital swelling, chemosis, proptosis, diplopia, and reduced visual acuity. This infection is often unilateral and not associated with papilledema, pupillary changes, or systemic toxic symptoms.[5] Superior orbital fissure syndrome is characterized by external ophthalmoplegia, exophthalmos, and Horner's syndrome or orbital apex syndrome (when visual impairment is present).[5] Patients with orbital apex syndrome classically present with visual loss and ophthalmoplegia that are out of proportion to and/or precede signs of orbital inflammation, such as periorbital edema, proptosis, and chemosis.[18]

CS syndrome may also be mistaken for CST because it is characterized by multiple cranial neuropathies, Horner's syndrome, and either hyperesthesia or hypoesthesia of the trigeminal nerve.[9,10] Oculosympathetic and oculoparasympathetic effects to the pupil may or may not be present in CS syndrome.[10] Specific CNs involved include the oculomotor nerve, trochlear nerve, the ophthalmic and maxillary divisions of the trigeminal nerve, and the abducens nerve. Multiple etiologies of CS syndrome exist, such as primary or metastatic neoplasia, granulomatous inflammation (ie, Tolosa-Hunt syndrome), cavernous carotid fistula, and cavernous carotid aneurysm, all of which lead to local compression of CS contents.[5,10] Multiple etiologies listed earlier may also contribute to CS syndrome and are not exclusive to CST.

Evaluation

Patients with CST may present with a multitude of symptoms, including headache (52%–90% of cases), pyrexia (90% of cases), nuchal rigidity (40% of cases), periorbital and facial hyperesthesia or hypoesthesia and swelling, chemosis, proptosis, photophobia, diplopia, painful extraocular movement, ophthalmoplegia (50%–80% of cases), papilledema, loss of corneal reflex, retinal venous distention, and vision loss (8%–15% of cases).[5,16,19] Severe cases often present with prominent symptoms of sepsis including tachycardia, emesis, hypotension, confusion, rigors, and even coma.[5] Pyrexia associated with CST is characterized by a "picket fence" pattern with high spikes followed by normothermia, suggestive of thrombophlebitis.[17] Symptoms are often initially present unilaterally; however, they may progress bilaterally without appropriate treatment as the infection and thrombosis spread through the interCSs. Special consideration is warranted in patients with the aforementioned etiologies and those who are immunocompromised secondary to conditions such as diabetes, long-term corticosteroid use, chronic alcohol or drug abuse, human immunodeficiency virus or acquired immunodeficiency syndrome, or bone marrow transplantation.[4]

There are no specific laboratory diagnostic tests for CST, but the test results may help establish the etiology, causative microorganism, or both. Patients are often found to have marked elevations in polymorphonuclear leukocytes, erythrocyte sedimentation rate, C-reactive protein, and D-dimer.[20–22] In cases of septic CST, identification of the causative microorganism can help guide and focus treatment. In most cases, cultures of the primary infective source, blood, and concurrent suppurations yield positive results.[5] A lumbar puncture may be an appropriate diagnostic test when concurrent signs of meningoencephalitis are present. When performed, 82% to 100% of patients will demonstrate an inflammatory profile indicative of bacterial infection; however, cultures are only positive in approximately 20% of cases.[5]

Imaging

Although a diagnosis of CST is often possible from clinical symptoms alone, improvements in imaging modalities have enabled early diagnosis and improved surveillance. Various venographic techniques of magnetic resonance (MR) and computed tomography (CT) imaging are used to evaluate for venous thrombi.[23] Two-dimensional, unenhanced time-of-flight MR venography is often employed; this technique exhibits excellent sensitivity to slow flow and diminished sensitivity to signal loss from saturation effects as compared to 3-dimensional counterparts.[23] Additionally, contrast-enhanced MR venography with elliptical centric view-ordering has been applied to improve visual resolution of small vessels and venous sinuses.[23,24] CT venography is often used due to being more readily available and the speed at which imaging can be completed. It is approximately 95% sensitive for detecting venous sinus thrombosis when multiplanar reformatted images are employed.[25] In fact, the detail of most CT venography images is superior to that of MR venography, albeit with some technical difficulties relating to reduction of bony artifact without losing contrast signal within the adjacent sinus.[23,25] This bony artifact is especially prominent in CST because the CS lies along the skull base. Additional imaging modalities, including nonenhanced T2 MR sequences where anomalies may be present within the venous flow voids and nonenhanced CT imaging where hyperdensities may be seen along the course of the sinuses, may be utilized in the absence of the more optimal imaging mentioned earlier.

Regardless of the imaging modality performed, both direct and indirect radiographic signs should be used to confirm the diagnosis.[5] Direct signs include expansion or filling defects within the CS. In several studies, bulging of the lateral wall was found to be the most frequent and sensitive sign associated with CST.[26,27] Asymmetric filling between the paired CSs confirmed the diagnosis.[28] Indirect signs of CST include narrowing or occlusion of the intracavernous portion of the internal carotid artery, reversal of normal venous flow at the skull base, increased dural enhancement at the CS border, exophthalmos, fat pad edema, or secondary thrombosis of the superior ophthalmic vein, or petrosal, sphenoparietal, or sigmoid sinus.[5,26,27] Additional indirect signs include venous dilation, restricted diffusion, or thickening of the superior ophthalmic veins.[5] Despite the signs listed earlier, the imaging findings may be normal early in the disease; therefore, empiric treatment is warranted when a high index of clinical suspicion exists.

Therapeutic Options

Although no randomized trial results exist to guide treatments for CST, likely due to the sparsity of the disease, many investigators have reported on their experience and expert opinions on optimal treatment algorithms. At present, patient stabilization, resuscitation, and management of the underlying etiology are paramount.[17] Management of the primary etiology may include appropriate antibiotics, anticoagulation, corticosteroids, and surgery.

Certainly, antibiotics have had the greatest impact on the prognosis of CST, especially because most cases are septic in nature and the associated mortality has been reduced in the antibiotic era.[3,17] Empiric antibiotic regimens should be initiated as soon as possible based on the most likely pathogens involved, which depends on the suspected infectious origin.[7] Appropriate empiric regimens include a third-generation cephalosporin (ie, ceftriaxone, cefotaxime, ceftazidime), metronidazole, and nafcillin or vancomycin.[5,7,29] Antibiotic therapy may be guided once culture and sensitivity testing results are available. Antifungals are rarely necessary unless the patient is immunocompromised or has a history of poorly controlled diabetes.[5] Antibiotics should be administered for a minimum duration of 3 to 4 weeks, consistent with the management of other intravascular infections, such as endotheliitis and suppurative phlebitis.[29]

Corticosteroids have been investigated for the treatment of CST; however, no consensus has been established. Proposed benefits include decreased orbital and CN inflammation as well as decreased vasogenic edema.[5] Weerasinghe and Lueck[3] performed a literature review in 2016 in 88 patients with CST and found no significant improvements associated with corticosteroid therapy among patients who achieved full recovery, survived with disability, or died. Additionally, Canhao and colleagues[30] reported on the use of corticosteroids in dural venous thrombosis and demonstrated no evidence of improvement to support steroid use. Nonetheless, steroids may prove useful in patients who have adrenal insufficiency or pituitary dysfunction secondary to the disease.[7] Ultimately, the potential benefits of corticosteroids must be weighed against their potential immunosuppressive effects and prothrombotic properties.[31]

Anticoagulation, a mainstay of dural venous sinus thrombosis, remains controversial in the treatment of CST; however, in their literature review of CST patients, Weerasinghe and Lueck[3] found that anticoagulated patients had a considerably greater rate of full recovery (53.6% vs 32%) and fewer deaths (12% vs 28%). Interestingly, those investigators did not see a clear difference in morbidity between anticoagulated (34%) and non-anticoagulated (40%) patients.[3] Southwick and colleagues[20] found that mortality decreased from 40% to 14%, and morbidity was decreased from 50% to 34% in patients who were anticoagulated. Levine and colleagues[32] found no significant decrease in mortality; however, those investigators did find a decrease in morbidity associated with anticoagulation. The proposed benefits of anticoagulation are the prevention of thrombus progression, inhibition of platelet function, anti-inflammatory properties, and the promotion of antibiotic penetration in the thrombus.[3,5,7,32] Multiple anticoagulants have been used, including unfractionated heparin and low–molecular weight heparin, as well as multiple oral anticoagulants. No clear consensus on the treatment regimen exists; however, the authors recommend starting with rapidly reversible unfractionated heparin in initial stages of CST, followed by conversion to a long-acting agent when the patient's condition is stable, similar to that proposed by Bhatia and Jones.[17] Finally, the duration of anticoagulation is variable and should be continued until radiographic clearance of the CST is achieved; this generally ranges from several weeks to months.[5,7,17,28,29]

Surgical intervention is not a typical treatment for CST. In the case of a septic etiology, surgical intervention should be aimed at source control through surgical incision and drainage of any potential sources.[7] In the case of neoplastic CS syndrome leading to thrombosis, surgery may be beneficial in removing the offending mass lesion. When surgical intervention is performed, endoscopy is commonly employed to drain the sphenoid, ethmoidal, maxillary, and frontal sinuses.[3,5] Mastoidectomies and craniotomies (for empyema or abscess) may be performed along with orbital decompression, ventricular shunt placement (for post-infectious hydrocephalus), or both.[3,5]

SUMMARY

CST is a rare, but potentially lethal, condition that involves the formation of a thrombus resulting in venous hypertension within the CS secondary to either septic or aseptic etiologies. Aseptic etiologies include surgery, trauma, thrombophilia, neoplasia, head and neck cancers, hormone-replacement therapies, or pregnancy. Septic etiologies include sinusitis, otitis, erysipelas, mastoiditis, and odontogenic and facial infections. Evaluation of the patient begins with the clinical examination, which is important in identifying the many signs and symptoms that overlap with other, more common conditions.

It is imperative that clinicians maintain a reasonable degree of suspicion and move forward with appropriate imaging in the form of MR and CT modalities. Once a diagnosis or a high degree of suspicion is present, it is paramount to initiate appropriate therapies, which may include antibiotics, anticoagulants, and surgery for source control. Corticosteroids must be utilized judiciously because no clear consensus benefit has been established; however, they may be a necessity in patients with pituitary dysfunction or adrenal insufficiency.

CLINICS CARE POINTS

- CST is a rare and potentially lethal subset of cerebral sinus thrombosis that must be quickly identified and treated to produce the best possible outcome for the patient.
- CST has both septic and aseptic etiologies and affected patients may present with headache, "picket-fence" pyrexia, meningitic symptoms, facial hyperesthesia or hypoesthesia, chemosis, proptosis, ophthalmoplegia, vision loss, confusion, altered mentation, or even coma.
- Imaging modalities, such as 2-dimensional unenhanced time-of-flight and enhanced with elliptical centric view-ordering MR imaging and enhanced CT venography, are used to help establish the diagnosis when clinical suspicion exists.
- Treatments for CST mainly focus around the source and offending pathology control, which may include the need for antibiotics, anticoagulation, and surgery. Corticosteroid therapy is controversial but used in some cases based on clinician experience and preference.

ACKNOWLEDGMENTS

The authors thank Paul H. Dressel BFA for preparation of the illustrations and Carrie Owens MSILS and Debra J. Zimmer for editorial support.

DISCLOSURE

Dr J.M. Davies reports consulting fees, payment, or honoraria for lectures, presentations, speakers' bureaus, article writing, or educational events from Medtronic and Rapid Medical; support for attending meetings and/or travel from Medtronic and Rapid Medical; patents planned, issued, or pending from QAS.ai; participation on a data safety monitoring board or advisory board for the National Institutes of Health National Institute of Neurological Disorders and Stroke Strokenet; and stock or stock options from Synchron, Cerebrotech, and QAS.ai. No disclosures were reported by the other authors.

FUNDING

Research Grants: NIH R21/R01, NSF SBIR, UB-CAT, Buffalo Translational Consortium, Cummings Foundation, nVidia, Google. Financial Interest: QAS.ai, Rist Neurovascular, Cerebrotech, Synchron, HyperionConsultant/Advisory Board: Medtronic, Microvention, Imperative CareNational

PI/Steering Committees: StrokeNET DSMB, EMBOLISE, SUCCESS.

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